

THE UNSEEN ALGORITHM: ADDRESSING SOCIETAL BIAS IN AI DATASETS

By: Rumaisa Siddiqui, Gavin Trimino, Lily Montero, Dhruuv Sharma

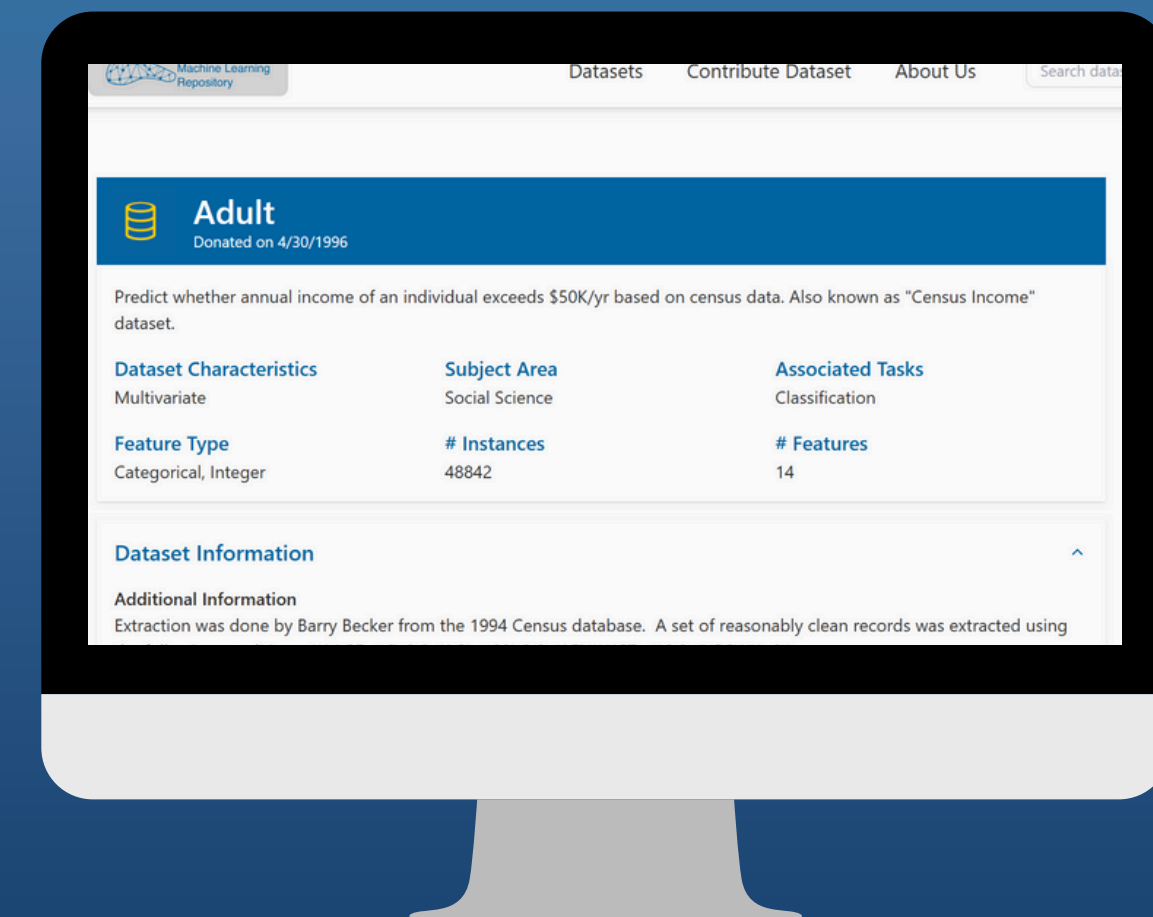


THE BIAS DATASET AT HAND

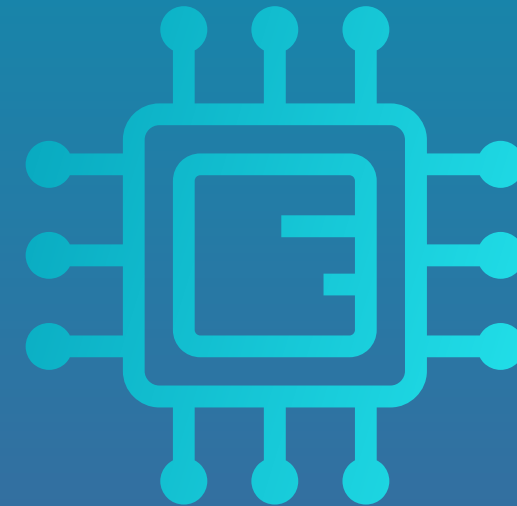
The UCI Adult Income dataset is a real U.S Census dataset from 1994. It contains 48,842 records about adults with features such as:

- Age, education, occupation, hours worked, race, sex, and marital status
- Target label: whether income is $> \$50K$ or $\leq \$50K$

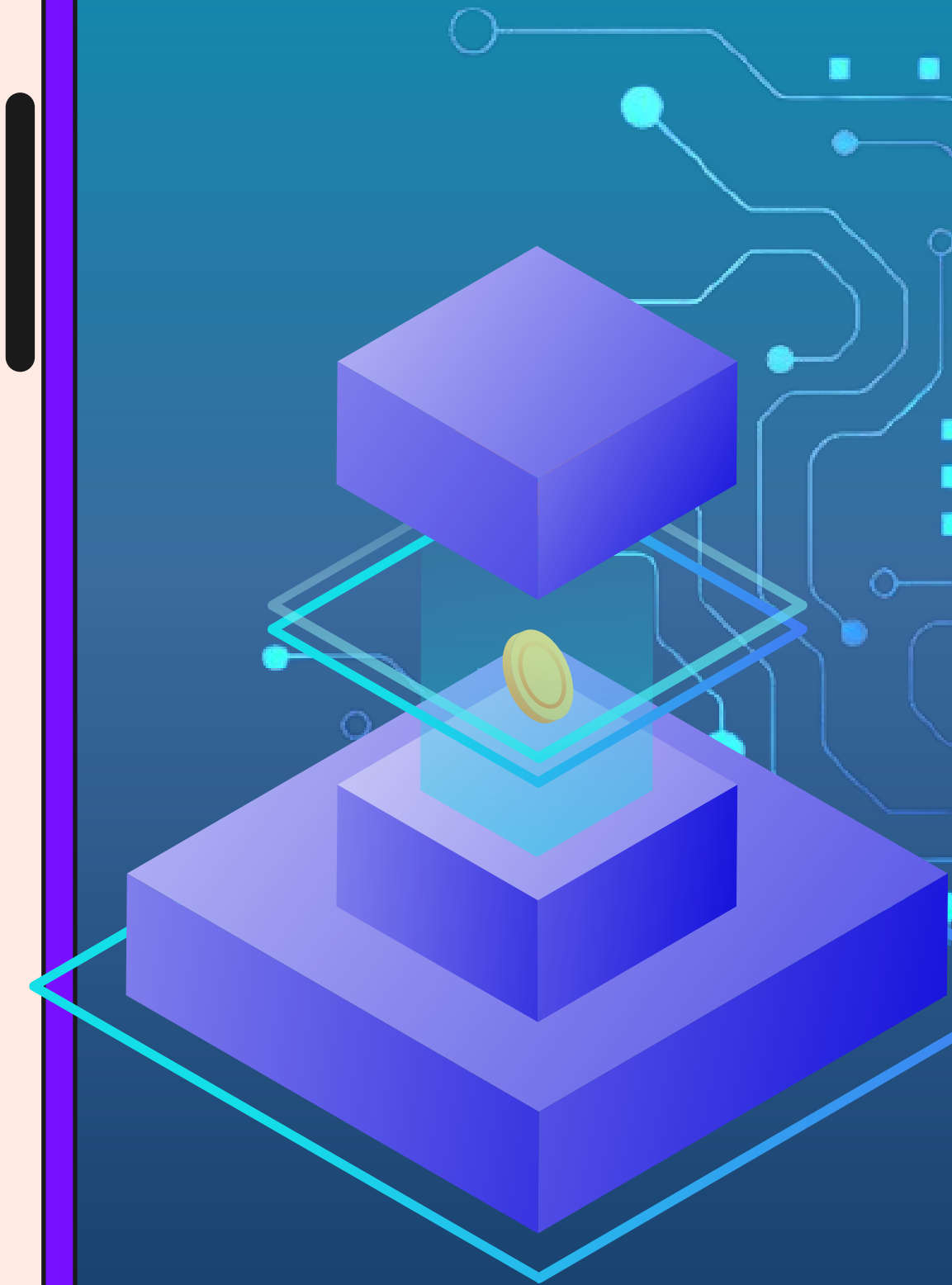
This dataset is widely used in AI research to study fairness and bias.



WHAT IS OUR GOAL?

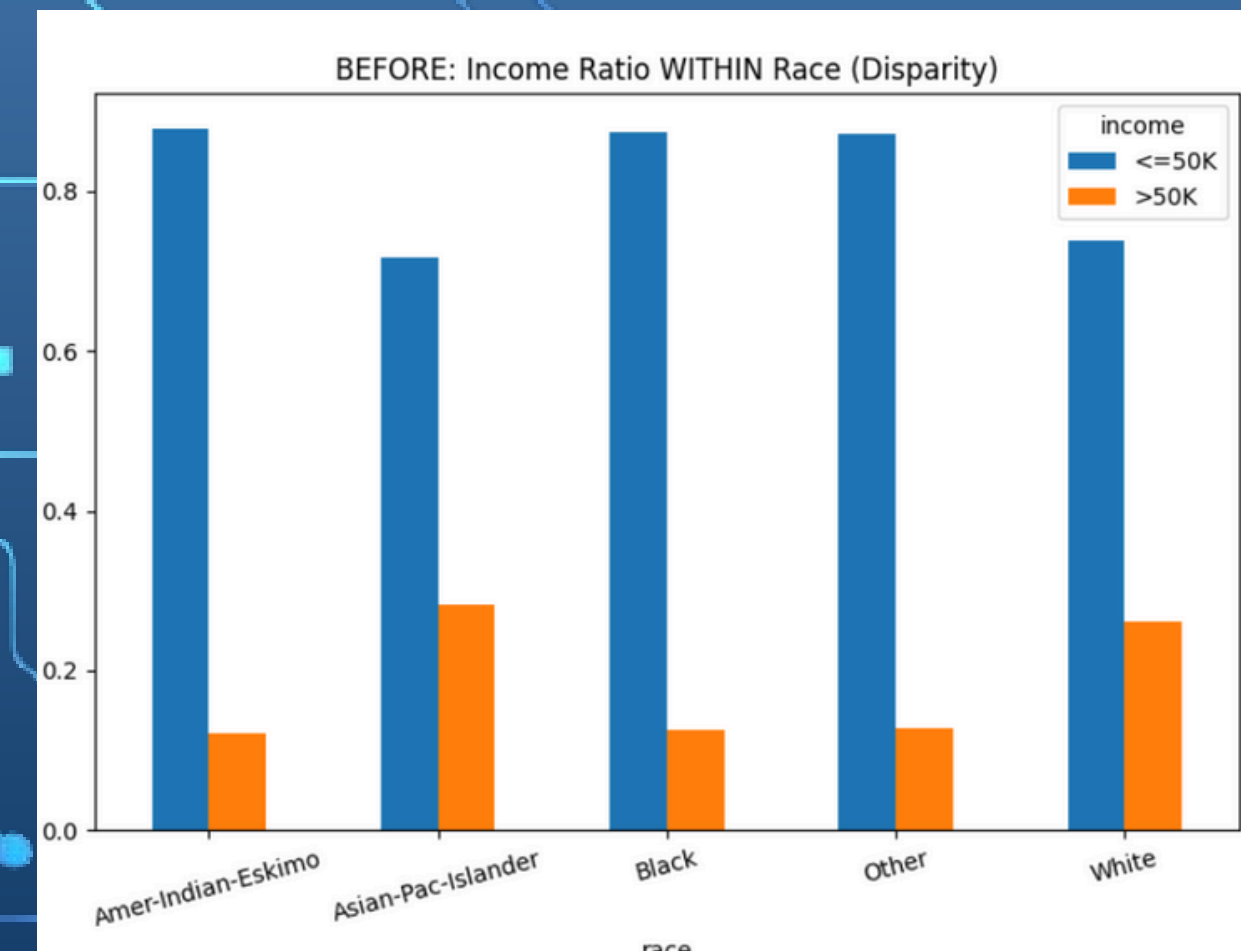
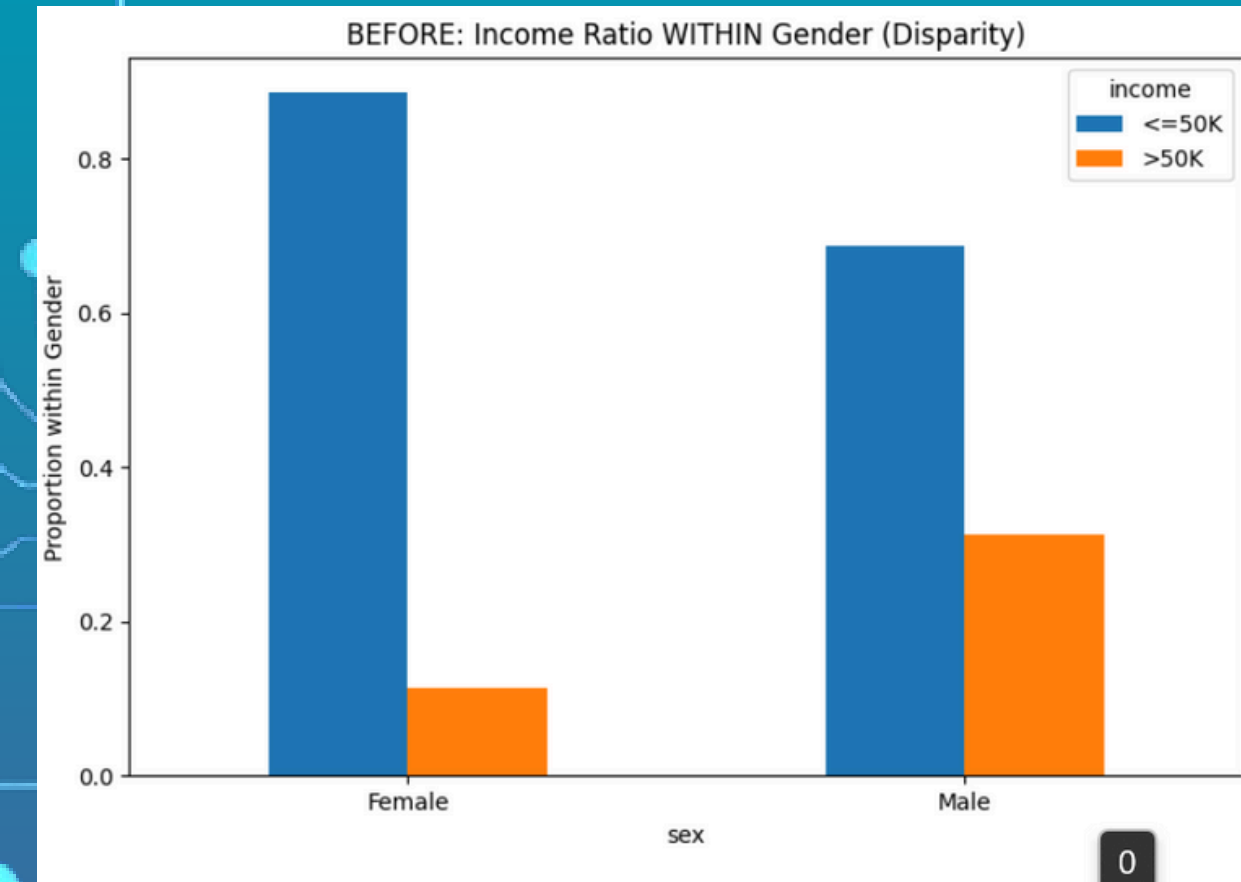


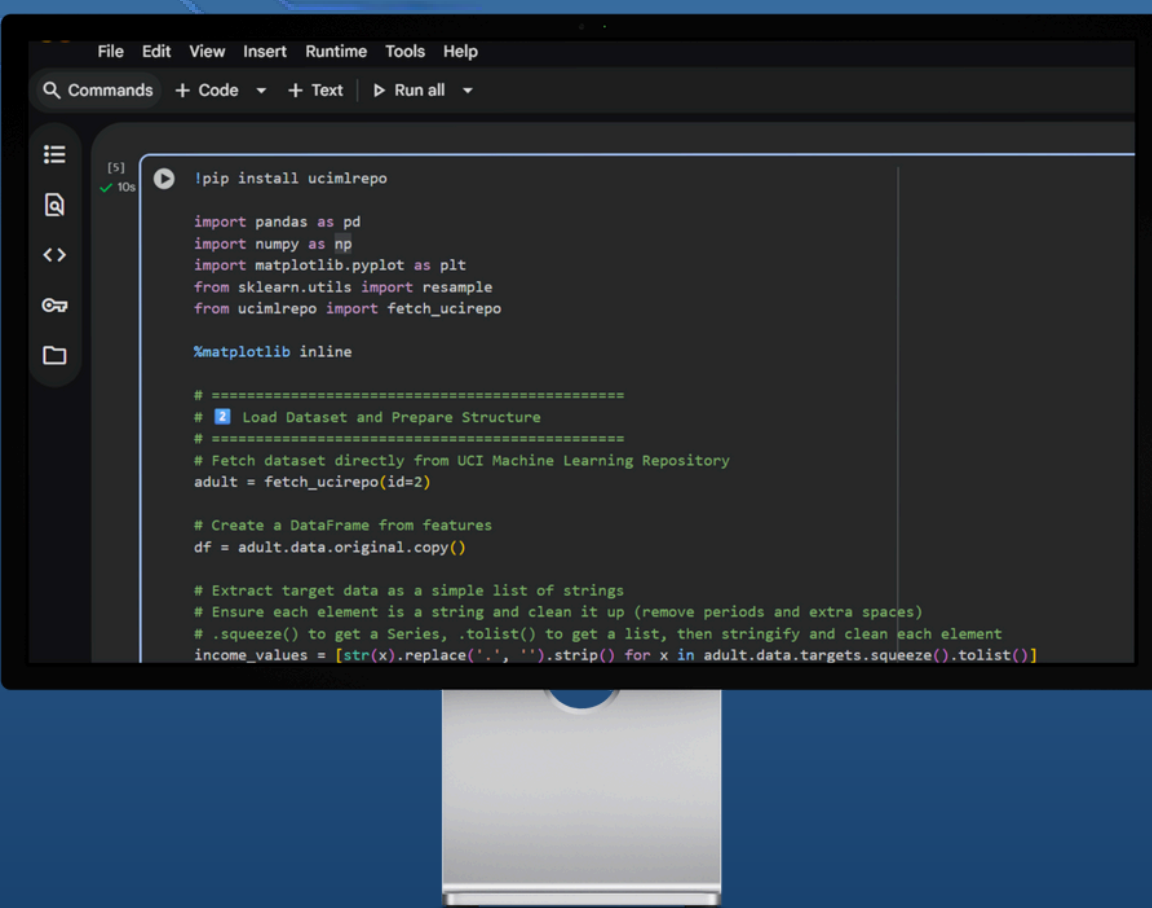
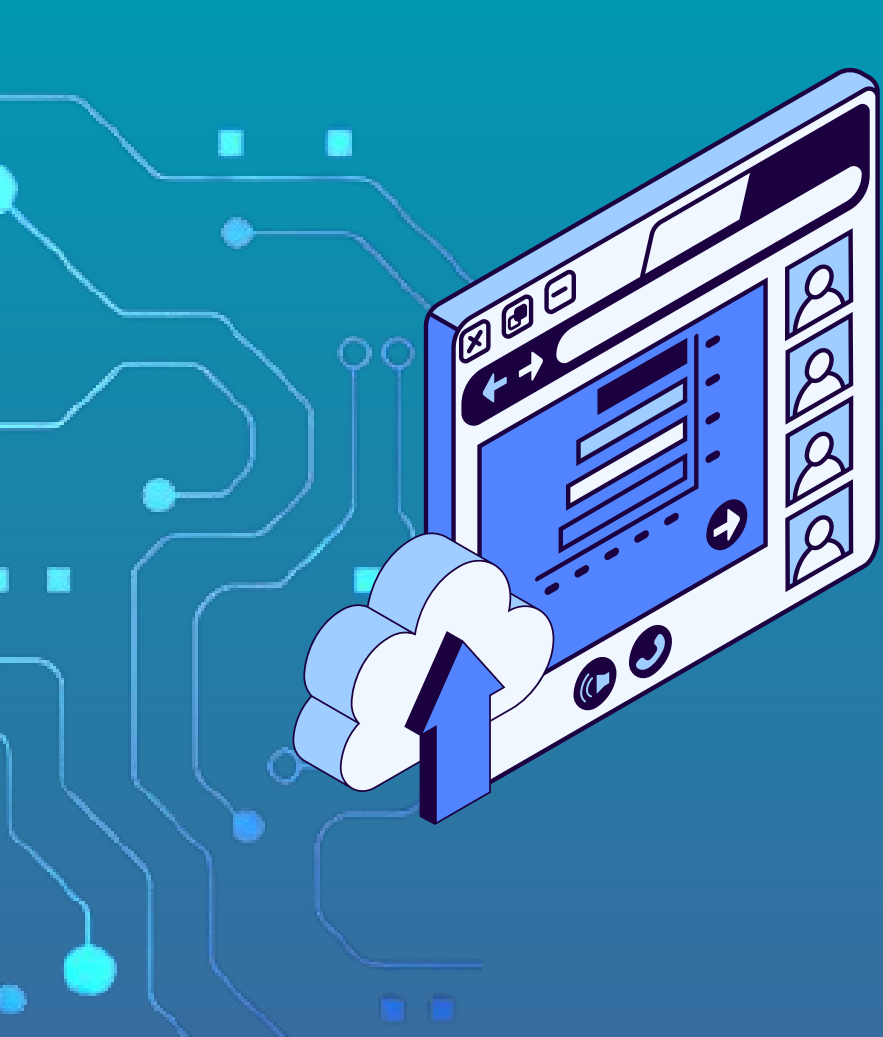
Our goal is to identify and reduce gender and racial bias in the UCI Adult Income dataset by improving the dataset's fairness, balance, and transparency. Instead of building a new model, we're improving the dataset itself, so that future AI models trained on it will make more equitable predictions.



WHAT IS THE PROBLEM?

AI bias occurs when models learn unfair patterns from unbalanced data. In the UCI 1994 income dataset, men and white individuals are overrepresented in high-income groups, which can cause AI to wrongly associate income with gender or race.





HOW WE USED AI TO REDUCE BIAS

We used Python and AI data processing libraries to detect and fix bias.

Here's the simplified process:

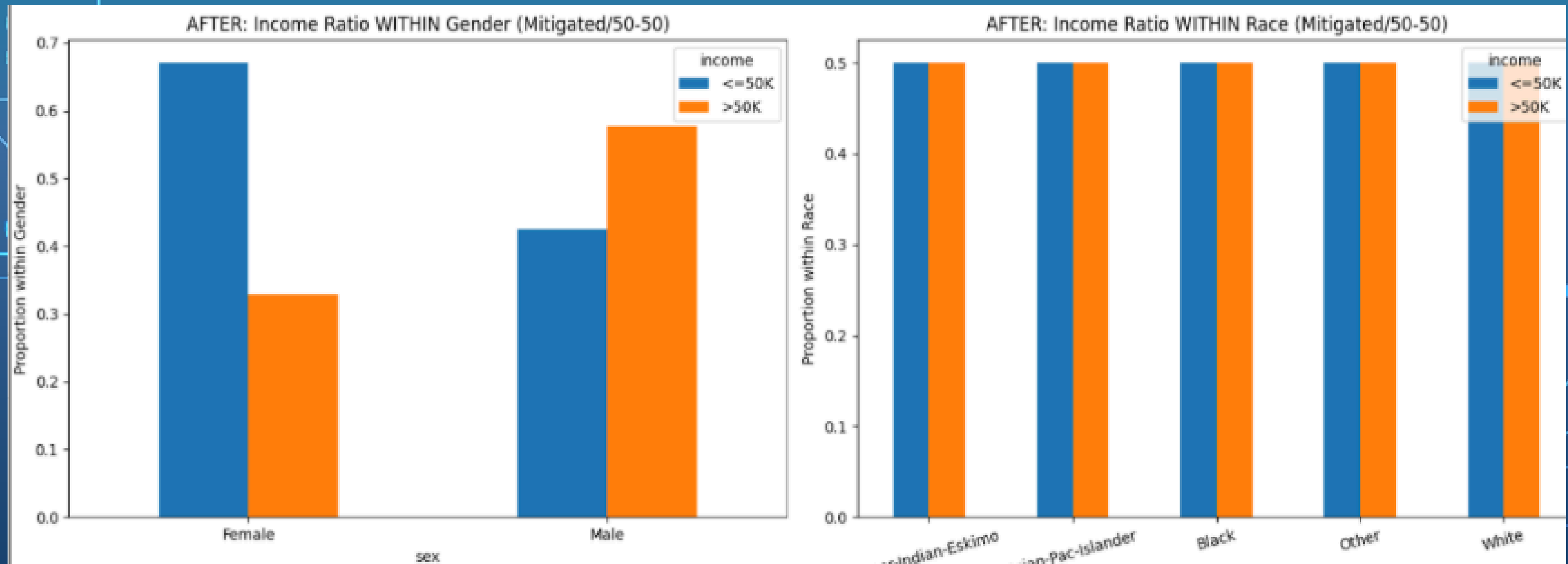
1. Analyze bias: We measured income distribution by gender and race.
2. Balance the data: We used an AI technique called **resampling** to give each group equal representation.
3. Verify fairness: After balancing, all groups had roughly equal income ratios.

```
df_balanced = resample(group, replace=True, n_samples=max_size)
```

This line tells the computer to “copy” underrepresented groups until everyone is equally represented.

AFTER FAIRNESS

Before balancing, men and certain races were more likely to appear in the high-income category. After balancing, every group had equal representation. This means any AI model trained on this new dataset would treat groups more fairly.

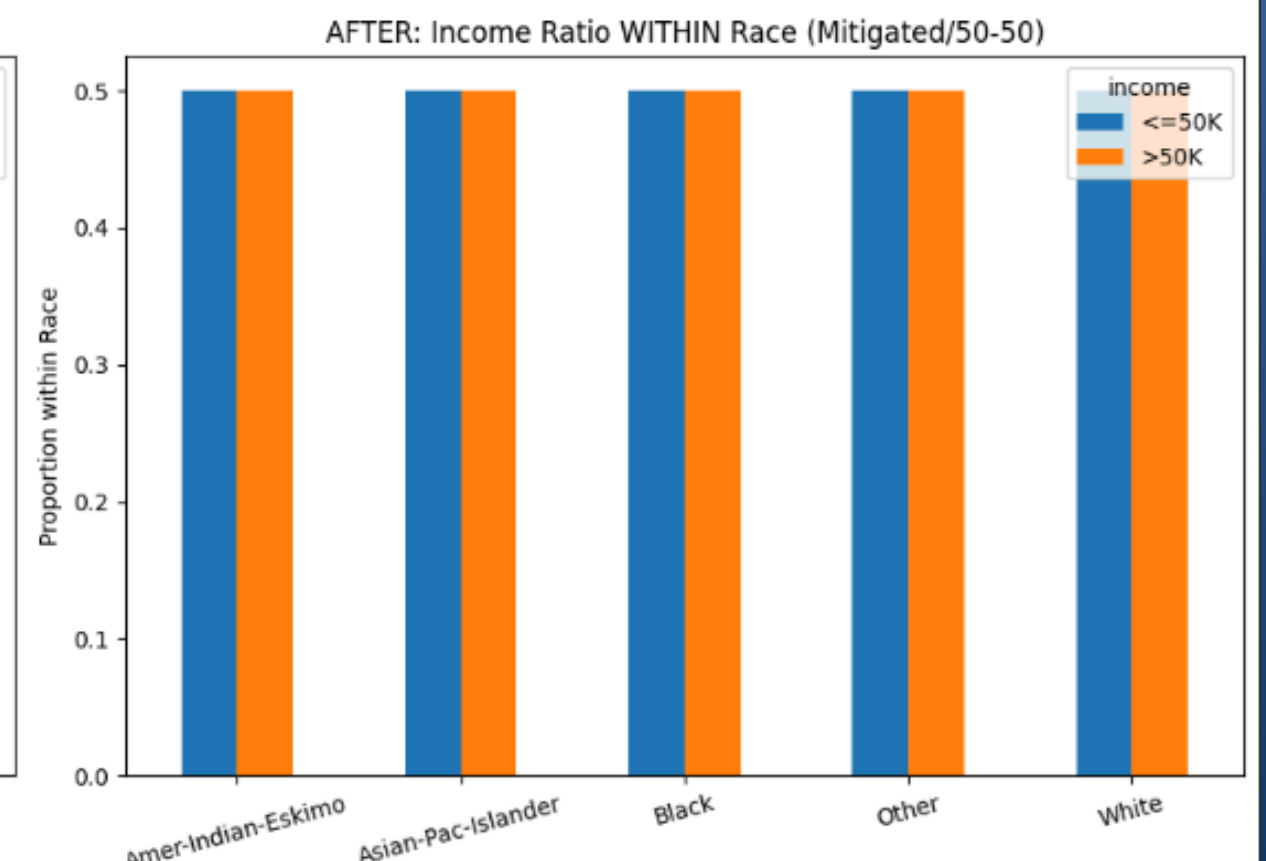
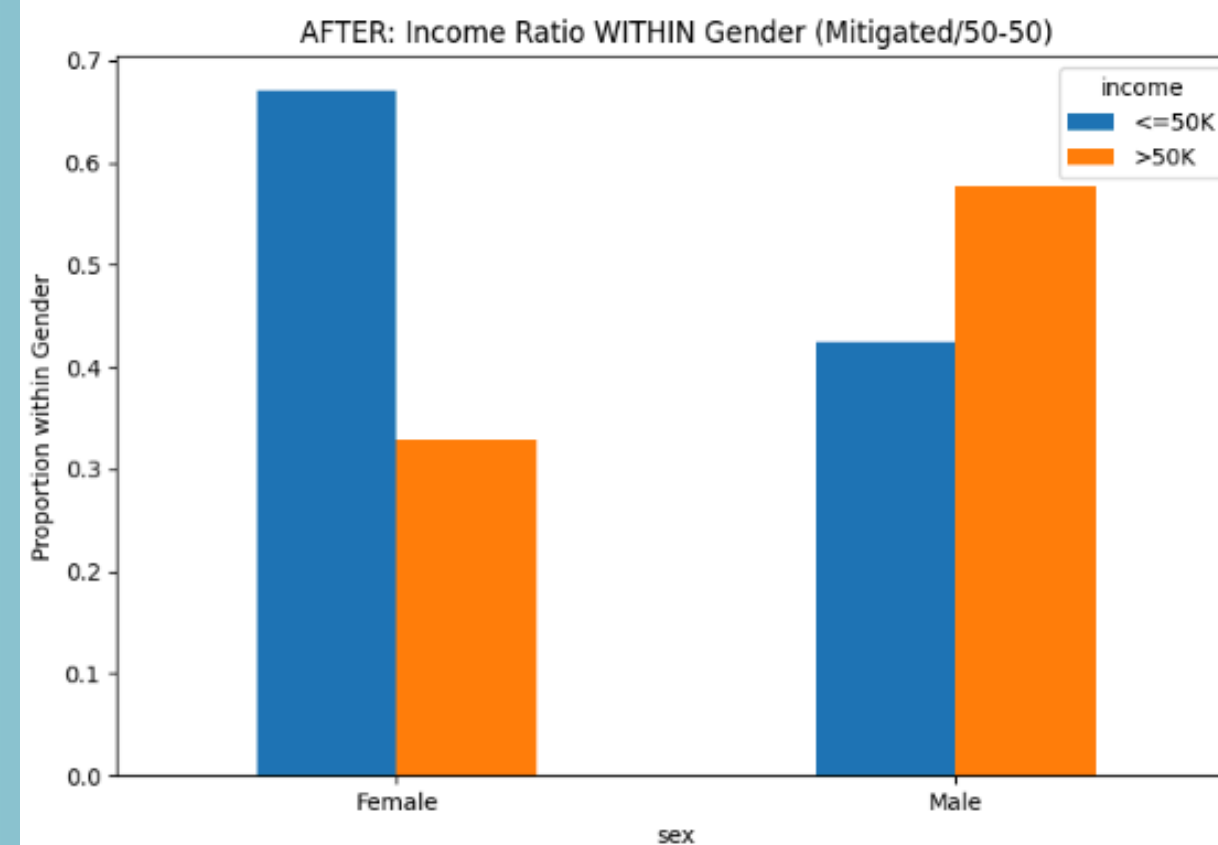
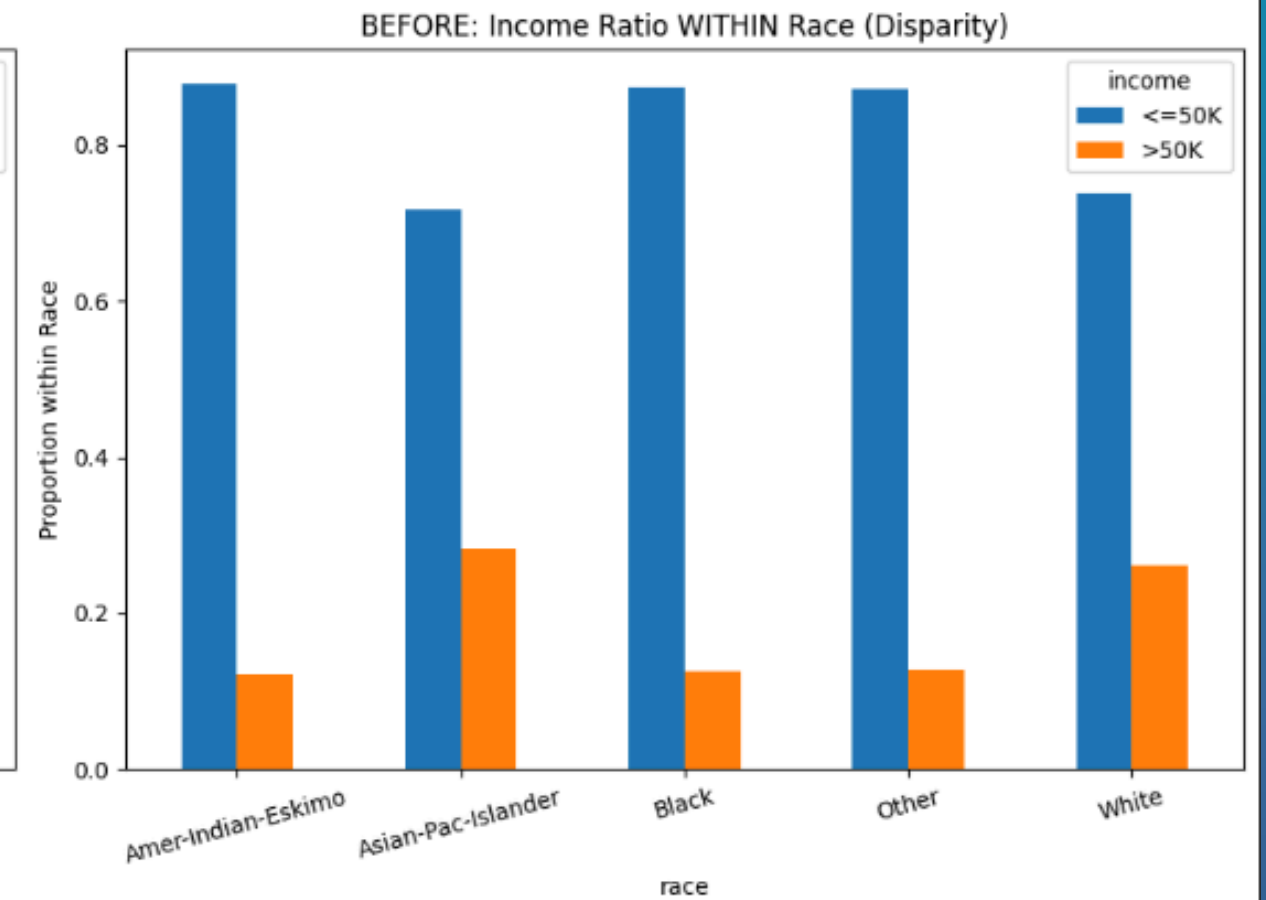
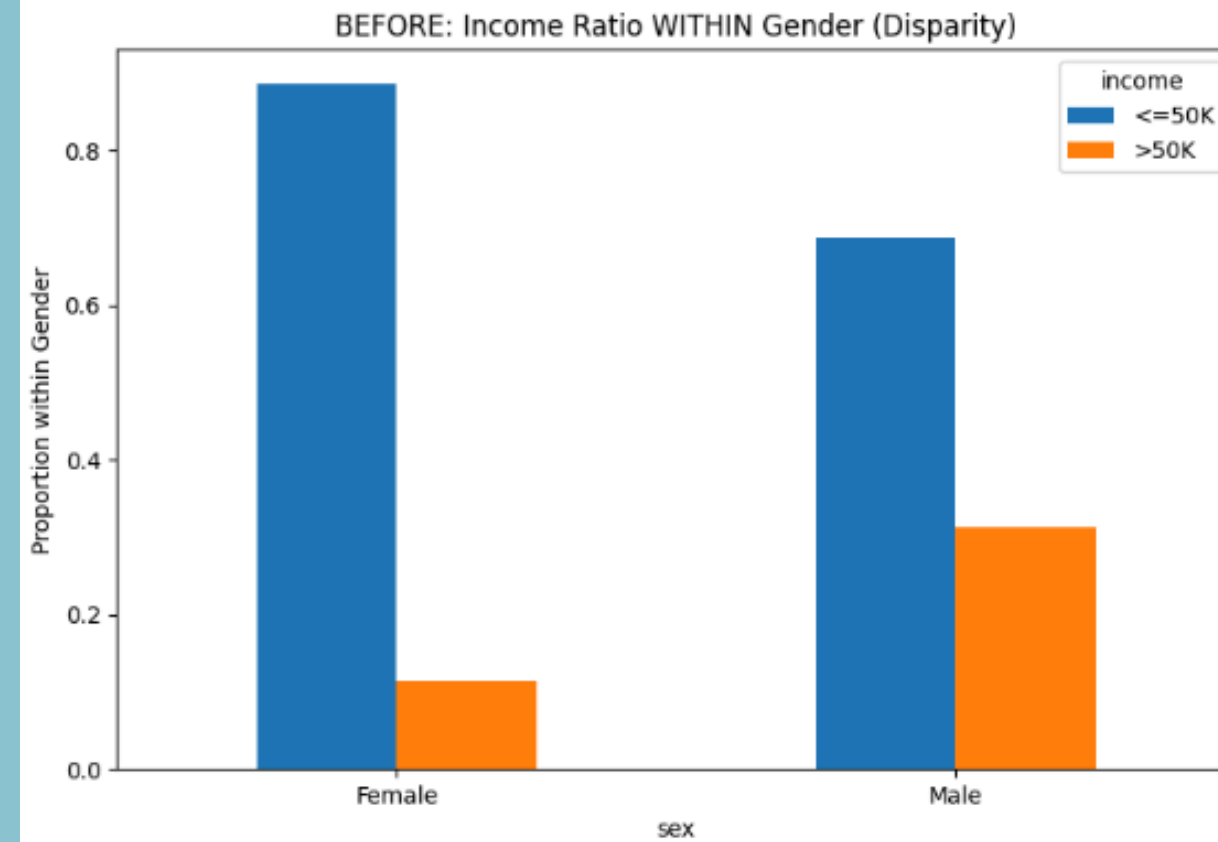


BEFORE VS. AFTER FAIRNESS COMPARISON

Awesome Web Browser



Bias Mitigation: Before vs. After Intersectional Upsampling





Most of our work was formulated through Gemini and ChatGPT, and our program was ran on Google Colab.

Our Progress



I need help to do the dataset and improve it

Perfect — that's the real hands-on part! Let's make this concrete.
To help you improve a dataset for **fairness, transparency, or privacy**, I'll walk you through **exactly what to do** — step by step — then we can tailor it to your specific data once you tell me what you're working with.

⚙️ Step-by-Step Plan to Improve a Dataset

Step 1. Pick a Dataset

If you don't already have one, start with a small, public dataset that's known to have bias issues or sensitive attributes.

Good beginner-friendly options:

- **UCI Adult Income Dataset** (gender & race bias)
- **COMPAS dataset** (criminal justice bias)
- **Credit Approval Dataset** (financial bias)
- **CelebA dataset** (facial attribute bias, if you're comfortable with images)

You can download these from Kaggle or UCI Machine Learning Repository.

+ Ask anything

ChatGPT can make mistakes. Check important info.

If I have a biased data set based on the UCI's Adult Income dataset , how would I fix the bias? Would I add more samples, or what?

That's a great and very practical question. Fixing bias in a dataset — like the **UCI Adult Income dataset** — isn't as simple as just adding more data; it depends on the *type of bias* and the *stage* of your project (data collection, preprocessing, or model training). Let's go step-by-step.

⚠️ Step 1: Identify What Kind of Bias You Have

Common biases in the **Adult Income dataset** include:

Type of Bias	Example	Description
Representation bias	Fewer female or minority examples	Some groups are underrepresented in the data.
Label bias	Income labels may reflect real-world discrimination	">50K" may be harder to reach for some groups due to structural inequities, not ability.
Measurement bias	Variables like "occupation" or "education" may not be measured equivalently across groups	Features don't represent the same real-world construct for everyone.
Sampling bias	Data comes from the 1994 U.S. Census	It may not reflect current population

Adult

Donated on 4/30/1996

Predict whether annual income of an individual ex

Dataset Characteristics

Multivariate

Feature Type

Categorical, Integer

Dataset Information

Additional Information

raction was done by Barry Becker from the 199

Install the ucimlrepo package

pip install ucimlrepo

Import the dataset into your code

from ucimlrepo import fetch_ucirepo

fetch dataset

adult = fetch_ucirepo(id=2)

data (as pandas dataframes)

X = adult.data.features

y = adult.data.targets

metadata

print(adult.metadata)

variable information

print(adult.variables)

DOWNLOAD (605.7 KB)

IMPORT IN PYTHON

CITE

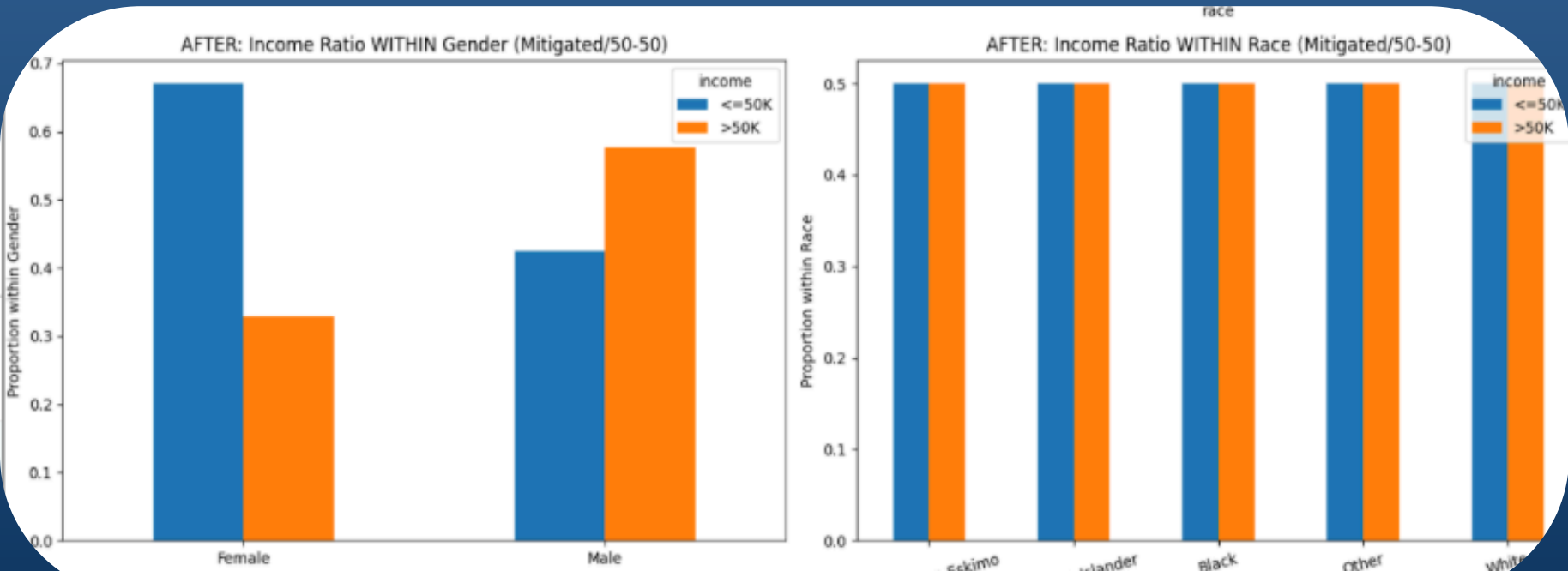
257 citations

545114 views

Keywords

census

Creators





DISCLAIMER



Dataset Disclaimer: Data Balancing Method

Method Used: Real Data Oversampling

- We achieved demographic balance by duplicating records of underrepresented groups.
- No Factual Data Altered:
 - No ages, incomes, attributes, or labels were changed.
 - No synthetic or fake data points were created.

Limitations and Purpose

- Inherent Bias: The dataset still reflects real-world historical biases present in the original data.
- Bias Reduction: Our method can reduce, but not eliminate, these inherent biases.
- Use Case: The results are strictly for educational purposes and are not suitable for real-world decision-making or deployment.



MODEL CARD- MAKING SURE OUR WORK HELPS, NOT HARMS

We wanted to make sure to make sure our approach is both ethical and beneficial:

- We preserve the integrity of the original dataset by keeping all factual data unchanged.
- We do not use or predict sensitive attributes like race or gender during model training.
- We're transparent about our process and limitations.
- We acknowledge that balancing does not remove every form of bias, but it is an important ethical step toward building fairer AI systems.
- The model accuracy percentage is 81.34%



CHALLENGES WE FACED



Some challenges we faced:

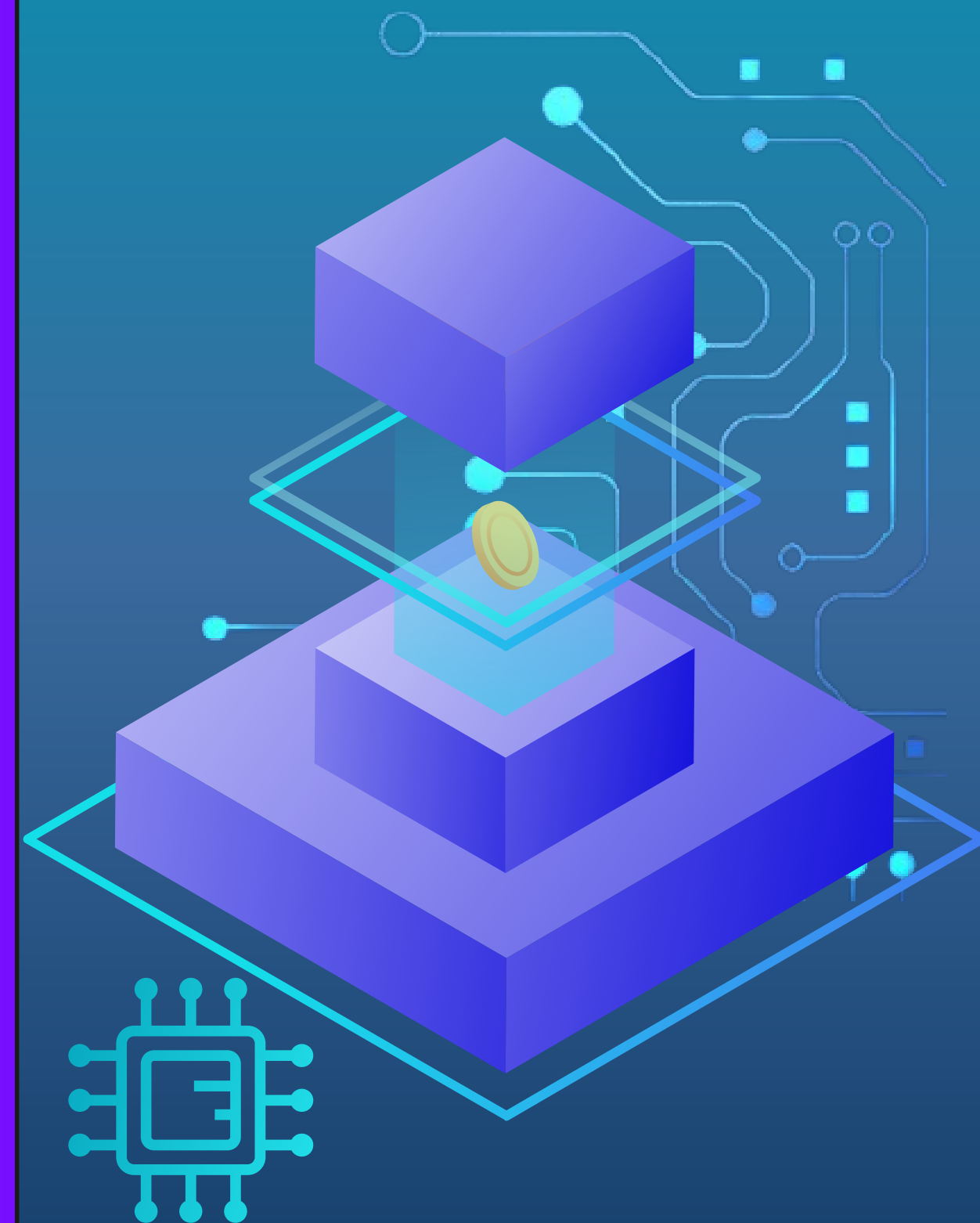
- Frequent syntax errors and small coding mistakes slowed us down.
- Our data-balancing attempts initially produced identical before/after graphs, so we had to revise the code.
- Gender bias persisted, showing that real-world datasets can be hard to fully fix.
- A more complex Race, Gender, and Income model would be needed for a complete solution.





Next Steps After Bootcamp

- Fix remaining gender imbalance
- Test model accuracy before vs. After bias mitigation
- Share the improved dataset with the AI student community
- Compare our balanced dataset to other unbalanced datasets
- Reach out to connections made during this program and explore opportunities
- Evaluate fairness using LLM models
- Explore more fairness methods





THANK YOU!

*Exploring the future of
computing together!*